GadPortfolio — UI/UX Design Document
My name is Gad Niyibigira.

App name: GadPortfolio
Course: SWE 3409 — Mobile Application Development
Institution: INES-Ruhengeri
Registration number: 25/26951
Verification code: MOB-A2-6951

1. Purpose and design direction

I built GadPortfolio as an Android portfolio app that a recruiter can install and understand within two minutes, without needing a network connection. It replaces a paper CV with something a recruiter can open on their own phone.

The design direction I chose is warm and editorial rather than corporate. Instead of the typical blue-and-white professional template, the app uses a warm off-white background (Paper), deep near-black text (Ink), and a terracotta accent (Ember), with a gold accent reserved for the verification code. Headings use Fraunces, a serif with personality; body text and interface controls use Geist, a clean modern sans. The Photography section shifts to a dark Ink background, giving it the feel of a gallery wing inside the rest of the app.

The design principles I worked from:

Identity first — the required "My name is..." sentence is the first content on the Home screen, overlaid on the hero.

Honest content — every skill, project, and photo shown is real.

Offline-first — no core feature requires a network connection.

Reusable components — cards, sections, buttons, and fields are shared across every screen.

Accessible by default — every interactive element has a label, contrast passes WCAG AA, and layouts reflow rather than clip when system text size is increased.

2. User flow and information architecture

Navigation structure:

The app uses a bottom tab navigator with five tabs, and a native stack navigator for the project detail screen.

text
Bottom Tabs
├── Home
├── Skills
├── Projects ──┐
├── Photography│
└── Profile   │
               │
Native Stack

└── ProjectDetail ◀── (pushed from Projects)

Flow 1 — First impression (Home):

Home shows the hero — my profile image with a dark scrim overlay, or a warm gradient when no image is set. On top of the hero, from bottom up: "My name is Gad Niyibigira.", the tagline "Just a Nerd.", my professional headline, and the verification code MOB-A2-6951 in a gold pill. Below the hero, three cards: About me, Primary skill, and What you will find here. If no profile image is set, an empty state invites me to add one from the Profile tab.

Flow 2 — Skills and education:

The Skills tab opens with a level filter row (All, Confident, Studied, Beginner) and a two-column card grid. Each card shows a brand icon or colored initial tile, a level badge, the skill name, and a category. Below the grid is a dashed "＋ Add a new skill" card that opens a bottom sheet for adding or editing. The Education & training section lists my three institutions.

Flow 3 — Projects:

The Projects tab lists my two genuine student projects. Each card shows a gradient accent strip, title, one-line summary, the first four technology chips, and a "View details ›" link. Tapping a card pushes the Project Detail screen onto the stack. Android Back returns to Projects and the tab session is preserved.

Flow 4 — Project Detail:

The detail screen shows the title, summary, and separate sections for Problem, My contribution, Technologies used, and Offline behaviour. This is the only stack screen; everything else is inside the tabs.

Flow 5 — Editing my profile:

The Profile tab has an image block (Choose from gallery / Take photo / Remove image), controlled text fields for headline, biography, and primary skill, single-choice availability chips, and Save / Reset buttons. On save, the app validates the fields and shows field-level error messages. Below the editor, an "About this app" section holds the app name, my full name, registration number, version, and a gold verification code card, followed by a short data-use statement.

Flow 6 — Adding a skill:

Tapping "＋ Add a new skill" opens a bottom sheet with a name field (minimum 2 characters), level chips, category chips, and an icon picker offering eight bundled brand icons or a colored initial tile. Saving adds the card to the grid immediately and persists it.

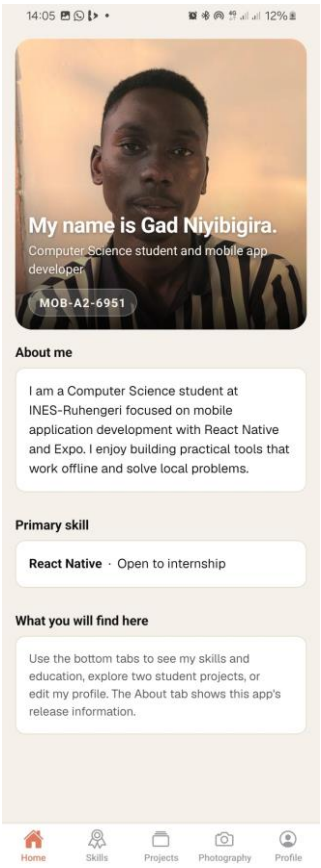
Flow 7 — Adding a photo:

Tapping "＋ Add a photo" in the Photography tab opens a dark bottom sheet. I can pick from the gallery or take a photo. The picked image is copied into a durable app-managed folder, then I add a title, description, and category. The photo appears in the masonry grid. A long-press removes it after a confirmation dialog.

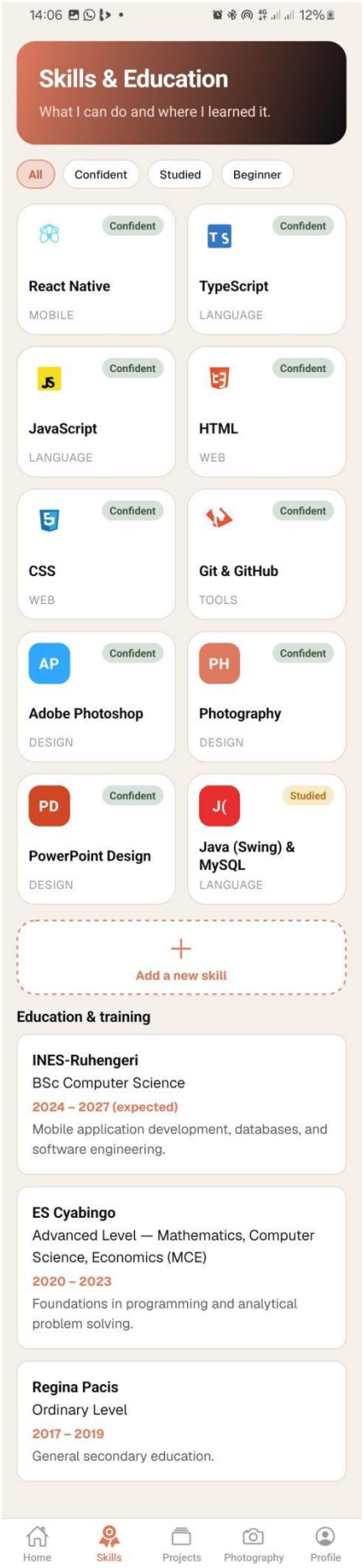
State and persistence:

All editable data — profile fields, the profile image, skills, skill categories, photos, and photo categories — is stored locally on the device with AsyncStorage. Images are copied into a durable app-managed directory so they survive app restarts. Nothing is uploaded to a server.

3. Annotated screens — Home and Skills



Home observations:
The hero shows my profile photo with a dark gradient scrim, on top of which sit the required identity sentence "My name is Gad Niyibigira.", the italic tagline "Just a Nerd.", my professional headline, and the verification code MOB-A2-6951 in a gold pill. The three cards below the hero present my biography, my primary skill, and a short orientation paragraph. The five bottom tabs are visible.

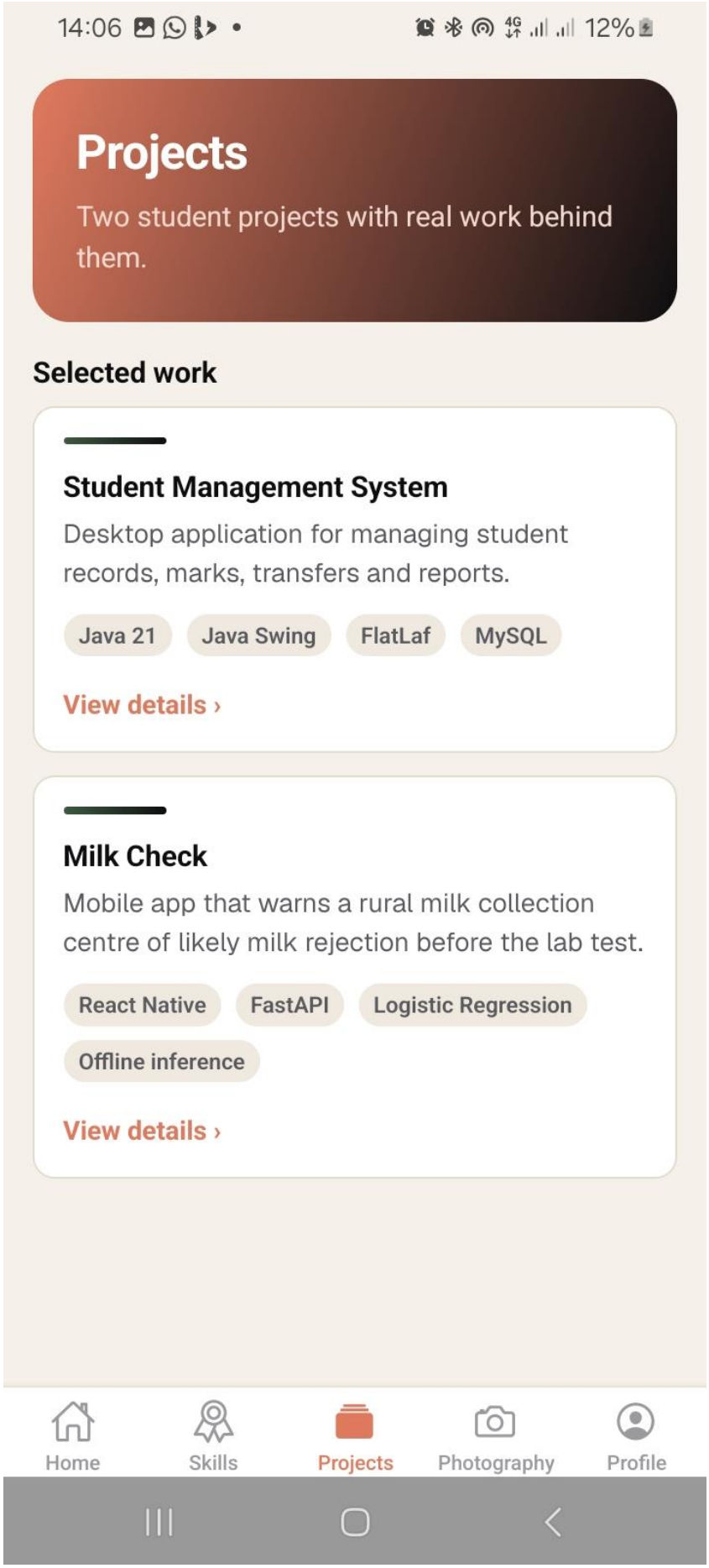

14:06
12%
Skills & Education
What I can do and where I learned it.
All
Confident
Studied
Beginner
Confident
React Native
MOBILE
Confident
TypeScript
LANGUAGE
Confident
JavaScript
LANGUAGE
Confident
HTML
WEB
Confident
CSS
WEB
Confident
Git & GitHub
TOOLS
AP
Confident
Adobe Photoshop
DESIGN
PH
Confident
Photography
DESIGN
PD
Confident
PowerPoint Design
DESIGN
J(
Studied
Java (Swing) & MySQL
LANGUAGE
Add a new skill
Education & training
INES-Ruhengeri
BSc Computer Science
2024 – 2027 (expected)
Mobile application development, databases, and software engineering.
ES Cyabingo
Advanced Level — Mathematics, Computer Science, Economics (MCE)
2020 – 2023
Foundations in programming and analytical problem solving.
Regina Pacis
Ordinary Level
2017 – 2019
General secondary education.
Home
Skills
Projects
Photography
Profile

Skills observations:
The Skills screen shows my genuine skills as cards in a two-column grid. Each card has a level badge (Confident / Studied) and a category label (Mobile, Language, Web, Tools, Design). Skills that have a bundled brand icon — React, TypeScript, JavaScript, HTML, CSS, Git — show that icon. Skills without a bundled icon — Adobe Photoshop, Photography, PowerPoint Design, Java (Swing) & MySQL — use a colored initial tile. Below the grid sits the dashed "＋ Add a new skill" card, then the Education & training section with three cards: INES-Ruhengeri (BSc Computer Science, 2024–2027), ES Cyabingo (Advanced Level MCE, 2020–2023), and Regina Pacis (Ordinary Level, 2017–2019).

Accessibility decisions on these screens:

Every card uses accessibilityRole="button" with a descriptive accessibilityLabel that includes the skill name, level, and category. Level badges use both a color and a text label so status is never conveyed by color alone. Section titles are marked with accessibilityRole="header". Touch targets on the tab bar and cards exceed 44px in both dimensions. The Home hero text uses a subtle text-shadow so it stays readable on top of any hero image the user picks.

4. Annotated screens — Projects and Photography

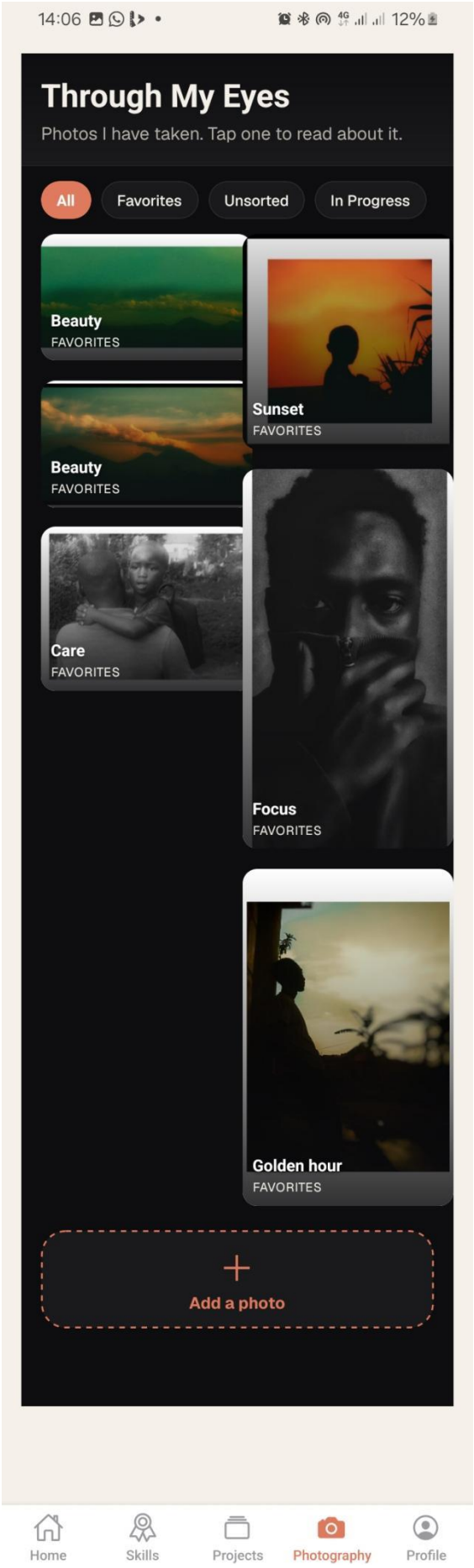

14:06
12%
Projects
Two student projects with real work behind them.
Selected work
Student Management System
Desktop application for managing student records, marks, transfers and reports.
Java 21
Java Swing
FlatLaf
MySQL
View details ›
Milk Check
Mobile app that warns a rural milk collection centre of likely milk rejection before the lab test.
React Native
FastAPI
Logistic Regression
Offline inference
View details ›
Home
Skills
Projects
Photography
Profile

Projects observations:

Two project cards are shown, both my own work:

Student Management System — a desktop application built in Java Swing and MySQL that manages student records, marks, transfers, and reports. My contribution covered student registration, update, delete, search, marks entry with automatic grade calculation, and the report generation with Excel export and printable QR ID cards. Technologies: Java 21, Java Swing, FlatLaf, MySQL, JDBC, Apache POI, ZXing, JavaMail.

Milk Check — a mobile app for a rural milk collection centre that warns collectors of a likely rejection before the lab test. I built the React Native app that records farmer ID, litres, temperature, and hours since milking, and shows a Low / Medium / High risk with a recommendation. The same logistic-regression model runs on the phone so the app keeps working offline. Technologies: React Native, FastAPI, Logistic Regression, Offline inference.


14:06
12%
Through My Eyes
Photos I have taken. Tap one to read about it.
All
Favorites
Unsorted
In Progress
Beauty
FAVORITES
Sunset
FAVORITES
Beauty
FAVORITES
Care
FAVORITES
Focus
FAVORITES
Golden hour
FAVORITES
Add a photo
Home
Skills
Projects
Photography
Profile

Photography observations:
The Photography tab uses a dark Ink background and a warm gold accent for "Through My Eyes" in Fraunces Bold. Below the title sits a horizontal row of category chips (All, Favorites, Unsorted, In Progress), then a mixed-ratio masonry grid. Photos keep their true aspect ratio: each tile's height is computed from the image's own width and height, clamped between 0.6× and 1.8× the column width so nothing becomes absurdly tall or short. Each tile has a dark gradient scrim at the bottom with the photo's title and category overlaid. An empty state invites the user to add a first photo, and a dashed "＋ Add a photo" card sits below the grid.

Project Detail screen (stack route):

Tapping any project card pushes a Project Detail screen. It shows the title, summary, the problem, my contribution, technology chips, and offline behaviour. I tested the Android Back gesture from this screen on the installed APK: it returns to the Projects tab and the session is preserved.

5. Visual system, accessibility, and final implementation

14:06 • 12%

Edit profile

Changes are saved on this device only.

Profile image



Choose from gallery

Take photo

Remove image

Professional headline

Headline

puter Science student and mobile app developer

Short biography

Biography

I am a Computer Science student at INES-Ruhengeri focused on mobile application development with React Native and Expo. I enjoy building practical tools that work offline and solve local problems.

196/400

Primary skill

Primary skill

React Native

Availability

Open to internship | Open to part-time | Open to freelance | Not available

Save profile

Reset profile

About this app

App name	GadPortfolio
Full name	Gad Niyibigira
Registration number	25/26951
Version	1.0.0

Verification code

MOB-A2-6951

GadPortfolio stores your profile text, skills, and photos only on this device. Nothing is uploaded to a server. Removing a photo, skill, or profile image deletes that data from this device. The app works offline; no network connection is required.

Home | Skills | Projects | Photography | Profile

Color palette:

Token	Hex	Usage
Paper	#F5F1EA	App background
Surface	#FFFFFF	Cards
Border	#E2DACB	Hairline separators
Ink	#0E0E10	Text; Photography background
Text muted	#5A5A60	Secondary text
Ember	#E07A5F	Primary accent (buttons, active tab)
Moss	#3D5A40	Secondary accent
Gold	#E9C46A	Verification code, tagline

Typography:
Fraunces Bold and SemiBold carry headings, section titles, skill names, and photo titles. Geist Regular, Medium, and Bold carry body text, buttons, tabs, and fields. Both fonts are SIL OFL licensed and ship with the APK. Headings use slightly negative letter-spacing so the serif reads as considered rather than decorative.

Spacing and layout:
Spacing follows a 4 / 8 / 12 / 16 / 24 / 32 px scale, defined once in the theme module. Cards use a 20px corner radius. Section headers use the Ember or Ink accent, never a decorative gradient. Lists and grids use flex: 1 and flexWrap: 'wrap' so content reflows on narrow screens rather than clipping.

Contrast:
Ink on Paper passes WCAG AAA (≈15:1). Muted text on Paper passes AA (≈7:1). White on Ember passes AA (≈4.6:1). Gold appears only on Ink backgrounds, where it exceeds AA. I checked these with the WCAG contrast formula before locking the palette.

Accessibility decisions:

Every interactive element has accessibilityRole and a descriptive accessibilityLabel.

Field-level error messages use accessibilityLiveRegion="polite" so screen readers announce them when they appear.

Minimum touch target of 44px. Buttons are 48px tall.

Tab labels stay visible at all times — no icon-only navigation.

Safe-area insets are respected at the top and bottom so nothing sits under a notch or the gesture bar.

Status is never conveyed by colour alone. Skill levels have both a badge colour and a text label. Validation errors show text, not just a red border.

Text reflows rather than clips when the system font size is increased, because layouts use flex wrapping rather than fixed heights.

Large-text behaviour:

On a narrow screen with large system text, cards grow taller rather than truncate. Section titles wrap. Multi-line text inputs expand. The tab bar keeps its labels. I verified this on a Samsung Galaxy A04e running Android 14.

Implementation comparison — screenshots vs final app:

The screenshots in this document are from the final installed APK, not mockups. During development I made four decisions that changed from my original plan:

About moved into Profile. The About information — app name, name, registration number, verification code, version, and data-use statement — was originally a sixth tab. I folded it into the bottom of the Profile screen. This kept the tab bar to five tabs (Home, Skills, Projects, Photography, Profile), which reads more comfortably on a phone, without losing any required content.

Photography became its own dark tab. Originally part of Projects. Giving it its own tab and a dark background makes the photos read as a gallery rather than a list item.

Skills became a card grid instead of a list. Brand icons and level badges give the screen a stronger visual hierarchy, and the "＋ Add a new skill" card lets the user maintain the list from inside the app.

Custom fonts were added. The first working build used the system font. Once the layout was stable I added Fraunces and Geist, which give the app the editorial tone described on page 1. Both fonts are bundled in the APK so they work offline.

Testing summary:

I installed the final APK on a Samsung Galaxy A04e running Android 14 and tested the following on the device itself: launch without Expo Go, profile save surviving an app restart, gallery image selection, camera capture, adding/editing/deleting a skill, adding a photo to the Photography tab, using the whole app in Airplane Mode, and the Android Back gesture from the Project Detail screen. All eight passed.